ZOO501 Final Current Solved

Exon Intron?

Both **introns** and **exons** are the genetic material. But the essential **difference between** the two is that one is made into protein and the other is not. DNA molecules are made up of small molecules that are called nucleic acids. An **exon** is a sequence of nucleic acids that are represented **in the** mRNA molecule.

How to avoid polyspermy?

o **prevent** the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to **avoid polyspermy**: a fast reaction, accomplished by an electric change in the egg plasma membrane, and a slower reaction, caused by the exocytosis of the cortical granules

Defect of neurulation?

Neural tube **defects** (NTDs) are severe congenital malformations affecting 1 in every 1000 pregnancies. 'Open' NTDs result from failure of primary **neurulation**as seen in anencephaly, myelomeningocele (open spina bifida) and craniorachischisis.

Meroblastic cleavage?

meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg. These cells form a blastoderm.

Fraternal and Identical twin?

Twins are two offspring produced by the same pregnancy. **Twins** can be either **monozygotic**('**identical**'), meaning that they develop from one zygote, which splits and forms two embryos, or dizygotic ('**fraternal**'), meaning that each **twin**develops from a separate egg and each egg is fertilized by its own sperm cell.

What is blastocyst?

Blastocyst Blastocyst: A thin-walled hollow structure in early embryonic development that contains a cluster of cells called the inner cell mass from which the embryo arises.

How acrosomal reaction initiate in sea urchin?

In the **sea urchin**, contact with egg jelly **initiates** the **acrosome reaction**, which is a calcium-mediated process. ... The **acrosomal** vesicle (green) fuses with the plasma membrane, releasing enzymes from the tip of the sperm that aid digestion of egg jelly.

How genetic material exchange in paramecium?

The micronucleus is the generative, or germline nucleus, containing the **genetic material** that is passed along from one generation to the next. **Paramecium** reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis.

What is fate of metencephalon?

The **fate** of the brain vesicles is determined by the combination of transcription factors. Signals from the secondary organizer, the anterior neural ridge and the mesencephalon–**metencephalon** boundary (isthmus), regulate expression of the transcription factors in the adjacent region and organize its **fate**.

Function of Placenta?

The **placenta** acts to provide oxygen and nutrients to the fetus, whilst removing carbon dioxide and other waste products. It metabolises a number of substances and can release metabolic products into maternal and/or fetal circulations.

What is campomelic dysplasia?

Campomelic dysplasia is a severe disorder that affects development of the skeleton, reproductive system, and other parts of the body. This condition is often life-threatening in the newborn period. The term "**campomelic**" comes from the Greek words for "bent limb."6 days ago

Name 4 level gene regulation?

?????

Two different pathways of neural crest cell? (10)

Neural crest cells that become the pigmentsynthesizing melanocytes migrate dorsolaterally into the ectoderm and continue on their way toward the ventral midline of the belly.

The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. **(Yehi Samjh aaya mazeed PPTS chk krain).**

Mygonesis (10)

Myogenesis: The Development of Muscle Paracrine factors instruct the Myotome cells to become muscles by inducing them to synthesize the MyoD protein. Factors from the surrounding environment induce the Pax3 transcription factor. In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD. MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family). The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes. MyoD also directly activates its own gene. While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell. Transfection of genes encoding any of these myogenic proteins into a wide range of cultured cells converts those cells into muscle

(Internte)

Myogenesis is the formation of **muscular** tissue, particularly during embryonic**development**. **Muscle** fibers form from the fusion of myoblasts into multi-nucleated fibers called myotubes. ... In this stage, calcium ions are critical for **development**.

Primary and Secondary neurulation?

In **primary** neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.
In **secondary** neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary

Cortical granule reaction?

Cortical granules are membrane-bound structures containing proteolytic enzymes. However, whereas each sperm contains one acrosomal vesicle, each sea urchin egg contains approximately 15,000 cortical granules

Embryonic Shield?

Also called **embryonic shield**. in the early **embryo** of mammals, the flattened inner cell mass that arises at the end of the blastocyst stage and from which the**embryo** begins to differentiate.

Slow block polyspermy?

The **slow block** to **polyspermy** begins within 10 seconds of fusion of the sperm and egg plasma membranes. A compound called inositol triphosphate (IP3) causes the release of Ca++ from intracellular stores in the egg endoplasmic reticulum.

Osteogenesis?

Osteogenesis imperfecta (OI), also known as brittle bone disease, is a group of genetic disorders that mainly affect the bones. It results in bones that break easily.

Types of Cleavage?

the entire cell is divided equally. Four major holoblastic **cleavage types** can be observed in general: radial, spiral, bilateral, and rotational. Egg cells that have larger quantities of yolk undergo meroblastic **cleavage** after fertilization, in which only a portion of the zygote undergoes**cleavage**.

Response from egg to sperm?

The responses of the egg to the sperm can be divided into "early" responses, which occur within seconds of the cortical reaction, and "late" responses, which take place several minutes after fertilization begins.

Blastocoel function?
The **blastocoel** probably serves two major **functions** in frog embryos: (1) it permits cell migration during gastrulation, and (2) it prevents the cells beneath it from interacting prematurely with the cells above it.

Tunicate gastrulation?
Tunicates, like sea urchins and vertebrates, follow the deuterostome pattern of **gastrulation** in which the blastopore becomes the anus. **Tunicate gastrulation** is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm

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